LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	39
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of artificial feel system and helicopter trim system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of artificial feel system and helicopter trim system.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS and Trim system schematics; Instructor supervision required for safety regarding hydraulic lines.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the artificial feel system?
(b) Why is a trim system necessary in a helicopter?
(c) How does the pilot's control input relate to the main rotor torque?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Purpose of Artificial Feel System • Components of AFCS • Hydraulic Boost Mechanism • Feedback Force Generation • Safety Valves and Limits	L & Ds	PPT, IB	15 min
(b)	• Helicopter Trim System Overview • Trim Wheel Operation • Trim Cable and Pulley Arrangement • Effect on Main Rotor Pitch • Trim Indicator Function	L & Ds	PPT, PDF	20 min
(c)	• Interaction between AFCS and Trim • Pilot Workload Reduction • System Failures and Contingencies • Maintenance Checks • Operational Procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the artificial feel system compensates for control forces.
(b) Describe the sequence of operation when adjusting the trim wheel.
(c) Identify two potential failure modes in the AFCS and their effects.

12. Home Assignment and Follow Up	Prepare a brief note on how the trim system settings change during different phases of flight (hover, climb, cruise) to be discussed in the next lesson on actuators.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of various types of actuators series and parallel and combine.	Refer to the master syllabus reference list	LP 40

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Captain, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min